

An End-to-End AI Agent for Kaggle Competitions

Where should external knowledge enter an agent's pipeline?

Author: Wei-Hao Huang · School: National Chengchi University, Department of Economics
PI: Dr. Tso-Jung Yen, Institute of Statistical Science, Academia Sinica · Summer Internship

1 · The question a competition can answer

A competition scores, from outside, the three things a human investigator normally supplies — **judgment** about the problem, an **error model** for the improvement claimed, and the **honesty** of the claim itself — so an agent's design decision here can be measured rather than argued.

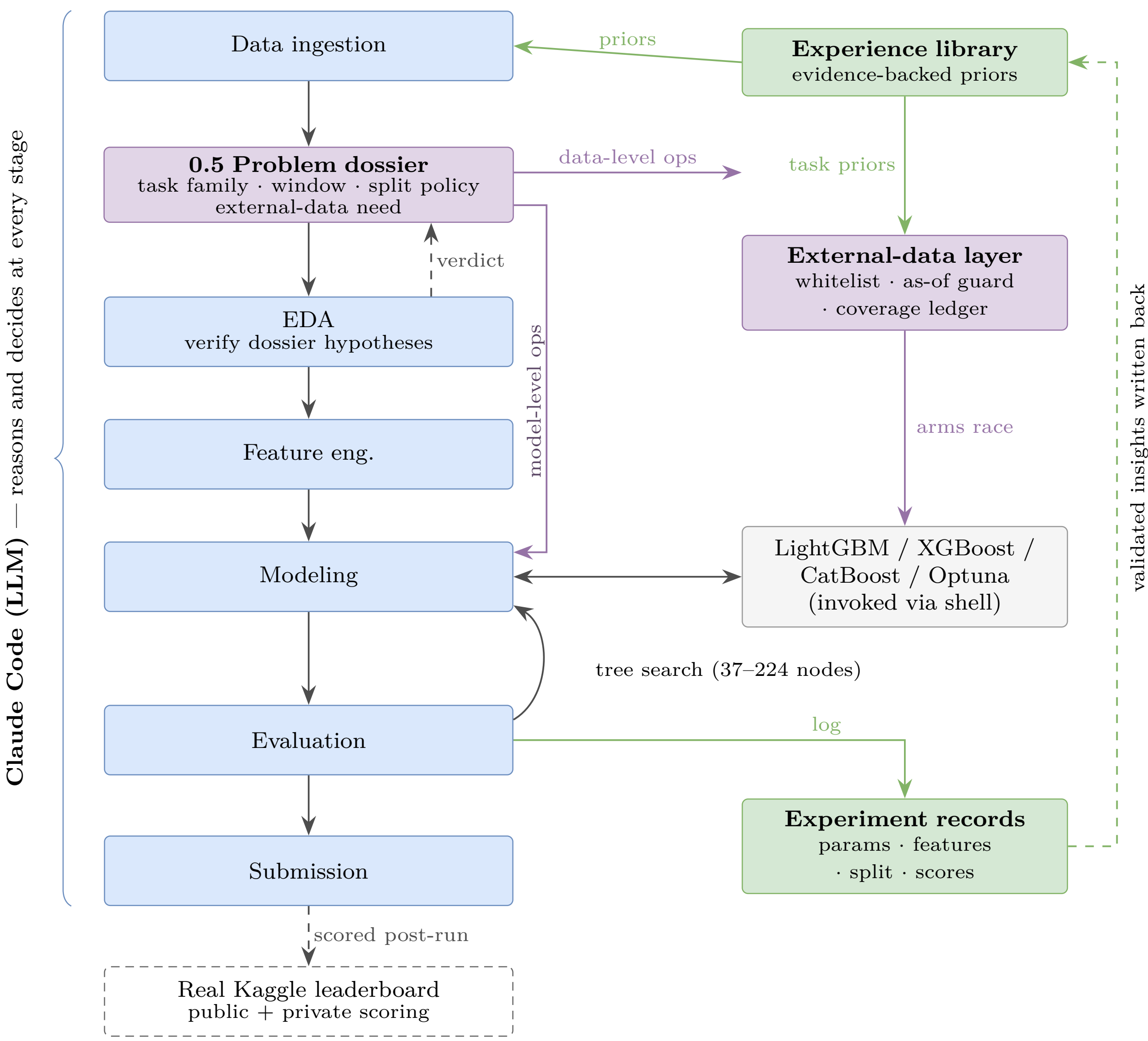
The scale is unforgiving: the tightest top-two private gap in the table in block 3 is **0.00003**, and rerunning an agent unchanged moves its score by a comparable amount — which is what makes both the judgment and its measurement hard.

Every experiment is logged, so every number here comes from a logged run. One question dominated the build:

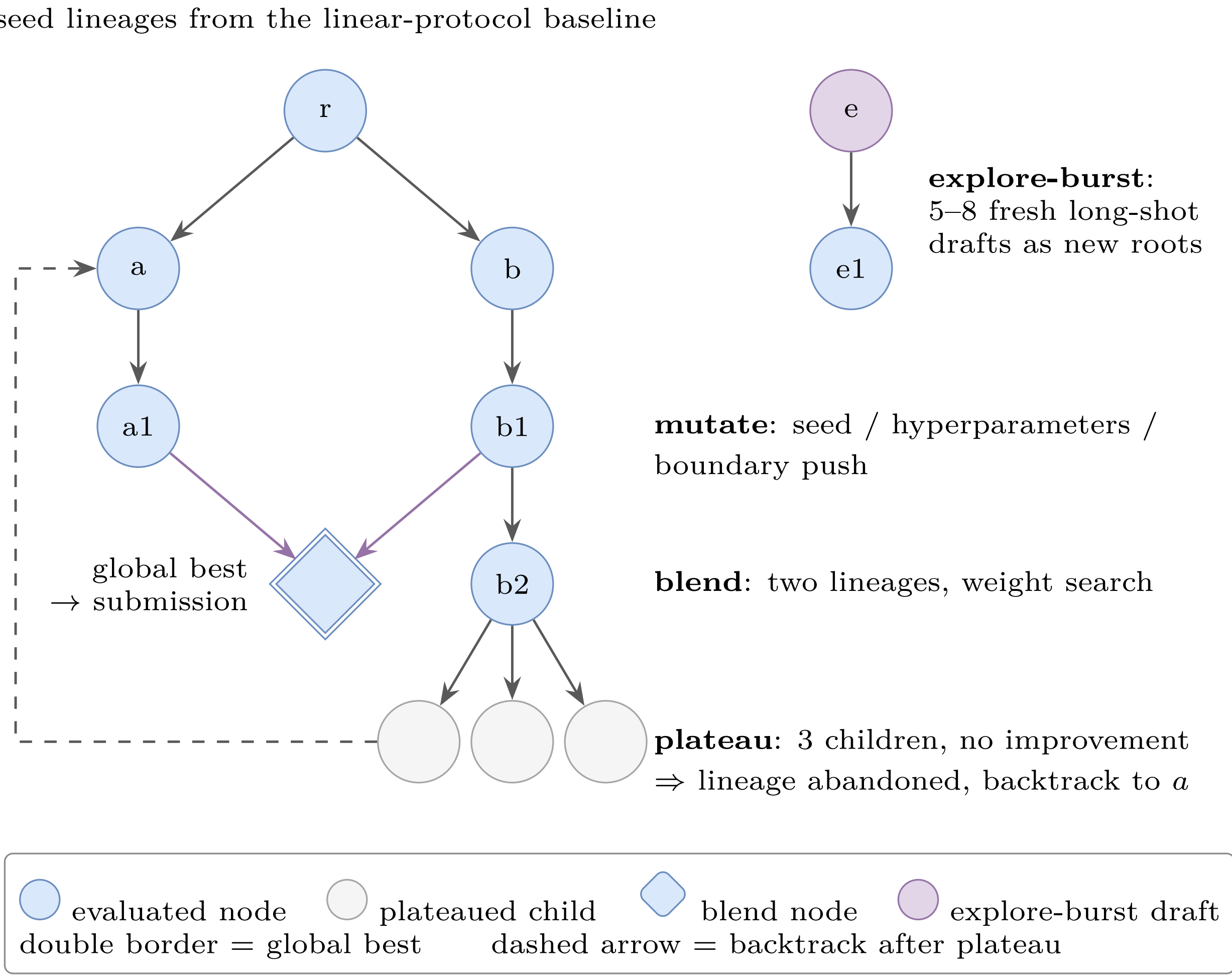
Where should external knowledge enter the pipeline?

2 · The design decision: knowledge enters upstream

The architecture has **one** design decision — where external knowledge enters — and both placements were measured: downstream injection, among the models being blended, moved scores by about 10^{-5} in our own implementation, so knowledge now enters **before any modelling**, at problem identification.



The LLM reasons and decides at every stage. **Purple** marks the injection layer: the problem dossier, the external-data module it feeds, and the typed operators it emits into the search. The leaderboard sits outside the agent.



One search tree. Every node is a complete, executable solution scored by cross-validation; a lineage is abandoned once its children stop improving; a **blend node over cached out-of-fold predictions** competes with single models and here becomes the global best that is submitted. 37–224 evaluated nodes per lane.

- In a blind sweep of the benchmark the layer flagged **exactly the competitions that need external data** and stayed silent on every other one.
- **Every** controlled external-data arm beat its matched baseline on the real leaderboard, each one paired-significant.
- **No local signal we tested selects the form of the injection.** The one surviving selector was killed by its first pre-registered prediction, so the layer races both forms inside the search.

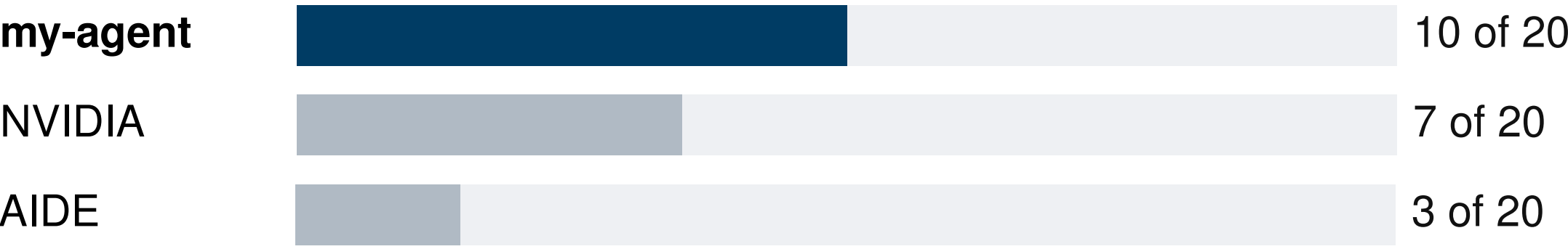
3 · Local validation cannot arbitrate

A claim about placement cannot be checked with the agent's own numbers: between agents no validation score settles anything — one lane's pooled out-of-fold SMAPE of **6.706** sat against a real leaderboard score of **49.40878**. Every verdict is a real submission, judged by an error model taken from Kaggle's own two-way split of the test set.

One submission gets a public and a private score; the shift between the halves is shared across agents and cancels in the between-agent gap, so a verdict needs the ordering to **replicate on both halves** and the gap to **exceed its own movement**. my-agent's benchmark lanes held no Kaggle credentials, so none could consult a leaderboard or submit; a separate operator step scored the frozen files. my-agent alone carries a cross-competition **experience library**; no ablation of it was run, so the standing cannot be attributed to it.

On that rule, 20 competitions × 3 pairings = **60 duels**, **all adjudicated: 38 decided, 22 undecidable** — not ties, but readings the test refuses to make. **my-agent takes 16–8 of its 24 decided duels (66.7%)**, against 11–15 of 26 (42.3%) for each frozen yardstick; the denominators differ because a duel is decided per pair.

Best private score — raw, direction-aware reading, no ties



Private-leaderboard scores, all 20 benchmark competitions

Comp.	Metric	my-agent	NVIDIA	AIDE	Comp.	Metric	my-agent	NVIDIA	AIDE
s3e1	RMSE ↓	0.55275	0.55550	0.55552	s5e1	MAPE ↓	0.08401	0.14594	0.17917
s3e3	AUC ↑	0.87157	0.89537	0.86863	s5e10	RMSE ↓	0.05576	0.05588	0.05579
s3e5	QWK ↑	0.57223	0.56974	0.58138	s6e1	Error ↓	8.71516	8.73239	8.72191
s3e7	AUC ↑	0.90441	0.90458	0.90109	s6e2	AUC ↑	0.95505	0.95508	0.95516
s3e9	RMSE ↓	12.33192	12.22913	12.30667	afsis	MCRMSE ↓	0.48917	0.69855	0.49861
s3e11	RMSLE ↓	0.29368	0.29624	0.29445	cat-in-dat	AUC ↑	0.80243	0.77084	0.80217
s3e14	MAE ↓	333.18611	330.71616	332.31556	conway	MAE ↓	0.10414	0.11189	0.11065
s3e16	MAE ↓	1.33822	1.34001	1.34224	tps-aug	AUC ↑	0.58930	0.58730	0.59106
s3e19	SMAPE ↓	49.40878	48.26558	48.34131	tps-jan	SMAPE ↓	6.29895	4.63651	10.66638
s4e11	Acc. ↑	0.94107	0.94076	0.94056	tps-sep	SMAPE ↓	28.49605	5.48351	9.72095

Private-leaderboard score of each frozen submission, printed as recorded; **bold** = best of the three. The frozen, credential-free re-run beats its own development-lane private score on **13 of 20**.

4 · Where it loses is where the design predicts

The standing says who won; where it loses says why: my-agent's two worst losses fall on the only two competitions here with a **mature, high-quality public solution** — NVIDIA's kernel reproduction dominates there, search wins the rest — and that mechanism, not the standing, is the result.

On tps-sep-2022 it scores **28.49605** SMAPE against 5.48351 for NVIDIA and 9.72095 for AIDE; its own public and private scores there (28.45070 / 28.49605) put the sampling noise two decimal places below the gap it loses by. On tps-jan-2022: 6.29895 against 4.63651.

The standing is ours to qualify too. Dropping s3e19 alone moves it from 66.7% to **72.7%**, and one duel sits **exactly** on its measured variance floor: the rule requires the gap to *exceed* the floor, compared in **exact decimal**, so it stays undecidable rather than a seventeenth win.

Expected	Experienced
<i>One run and one score decide a winner.</i>	22 of the 60 duels are undecidable: 19 where the ordering or the gap does not survive the move between the two halves of the test set, 3 on a between-run variance floor.
<i>The agent follows the rules by default.</i>	The audit discarded a contaminated lane whose tainted local score looked <i>better</i> than its clean replacement's (6.706 clean against 5.740 tainted); later, one lane's write-back polluted the next lane's priors — killed one minute in, library reverted, relaunched clean.
<i>The auditor is the thing that catches you.</i>	Re-reading the transcripts caught our own auditor exempting the commands every lane actually issues, so the one command it existed to catch was never examined. Corrected, a re-run over all 20 transcripts returns zero network findings.

5 · What transfers

Judgment, measurement and honesty were the three things under test; the placement result, the paired test and the audit answered them in that order. What transfers is not the ranking but the discipline that produced it: **placement of knowledge is an engineering choice with a measurable answer**, and a ranking published without an error model is the author's own noise.

- **State the error model with every claim.** A third of our own duels cannot be called, and the winner moves with the composition of the evaluation set.
- **Freeze the yardsticks.** Improving a yardstick mid-study makes earlier and later data incomparable, so neither was touched, even where we found weaknesses.
- **Isolate mechanically, then read the transcripts — including your auditor's.** Sandboxes stop the leaks you anticipated; reading what it actually did catches the ones you didn't.
- **Placement, not amount.** The dossier decides *whether* to inject; nothing local decides *how*, so both forms race inside the search.

github.com/930731haohao-aik/kaggle-agent